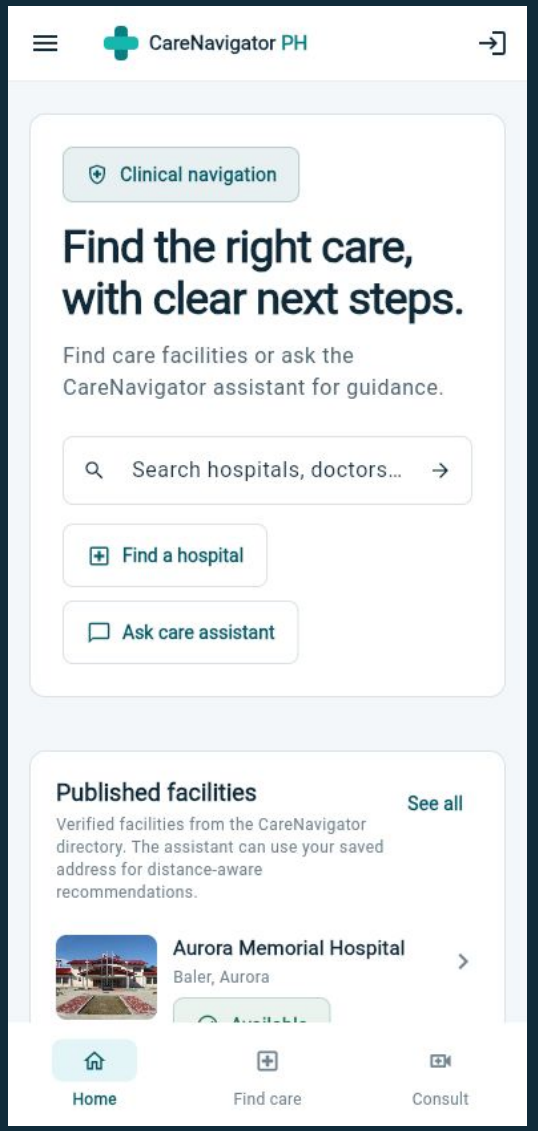
CARENAVIGATOR PH

System Demonstration and Defense

A role-aware platform for finding care, coordinating consultations, and managing clinical operations.

FLUTTER + SUPABASE

MOBILE + WEB



390 x 844 mobile viewport

Source: README.md; submission_assets/live_app_home_mobile.png

PROBLEM AND RESPONSE

One platform replaces a fragmented care journey

Without coordination

- Patients search across disconnected channels for facilities, clinicians, and availability.
- Consultation requests, medical files, messages, and follow-ups are handled in separate workflows.
- Hospitals and administrators need controlled, auditable access to operational and clinical data.

With CareNavigator PH

- Verified hospital and clinician discovery
- Deterministic emergency screening
- Guest and registered-patient consultation flows
- Role-specific clinical and operational workspaces
- Realtime communication with protected records

Source: README.md; SYSTEM_ARCHITECTURE.md sections 1 and 5

USERS AND SCOPE

Five roles share one governed care platform

Guest

Discover care, run preliminary screening, and submit a verified consultation request.

Patient

Book, reschedule, consult, message, and access records, prescriptions, labs, and files.

Doctor

Manage schedules, assigned patients, consultations, prescriptions, and clinical review.

Hospital admin

Manage staff, services, departments, capacity, rooms, ER status, and reports.

Super admin

Approve hospitals, govern accounts and permissions, review analytics, and audit activity.

Source: PRE_BUILD_REPORT.md - Role coverage

END-TO-END WORKFLOW

The demonstration follows a complete patient journey

PUBLIC

CLINICAL

CONTINUITY

Discover

01

Search verified hospitals, clinicians, services, and current availability.

Assess

02

Apply emergency rules first, then generate preliminary guidance.

Book

03

Submit guest intake or reserve an available slot as a registered patient.

Consult

04

Use secure messaging and approved consultation channels for online care.

Continue care

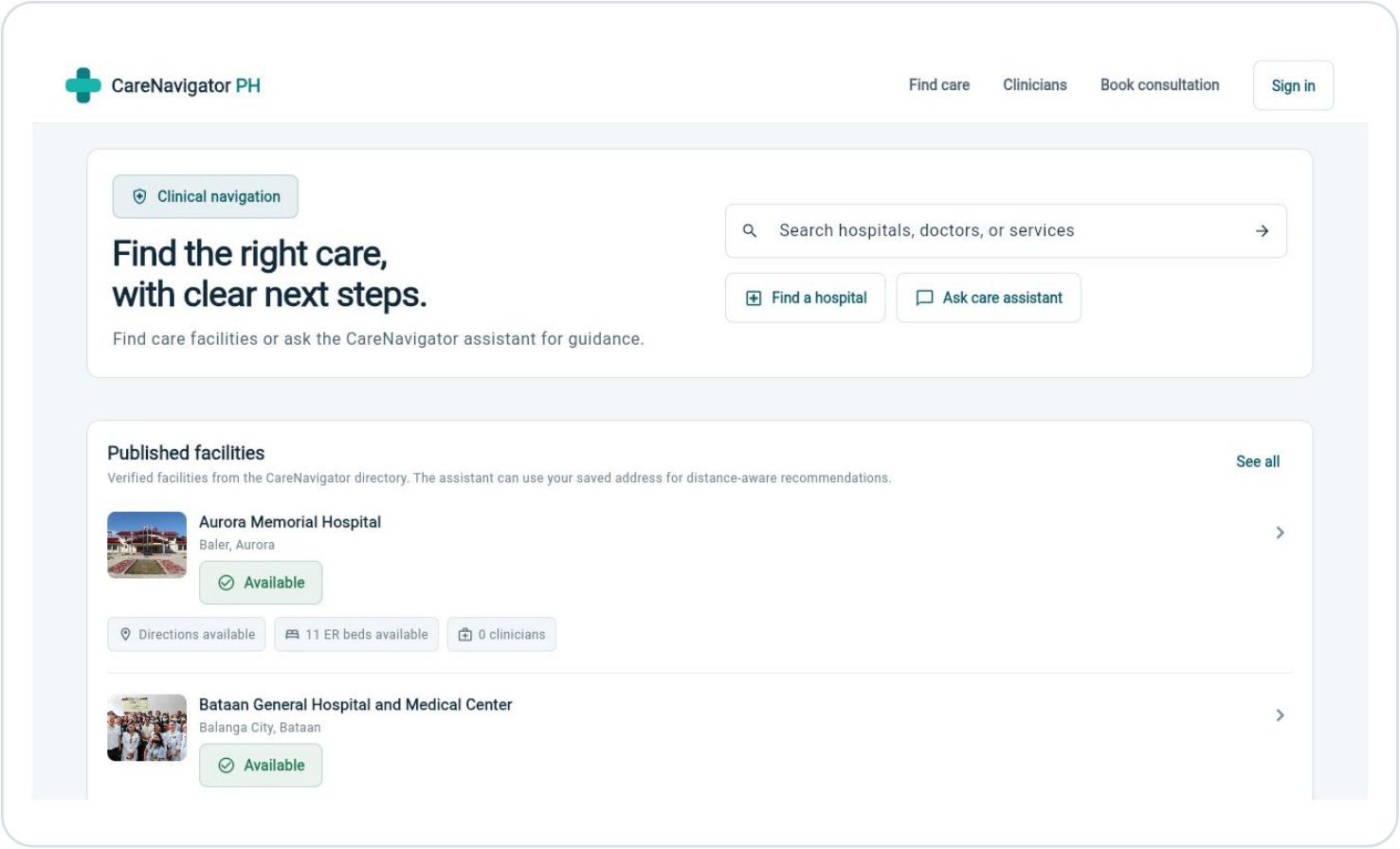
05

Receive prescriptions, lab requests, results, notifications, and follow-ups.

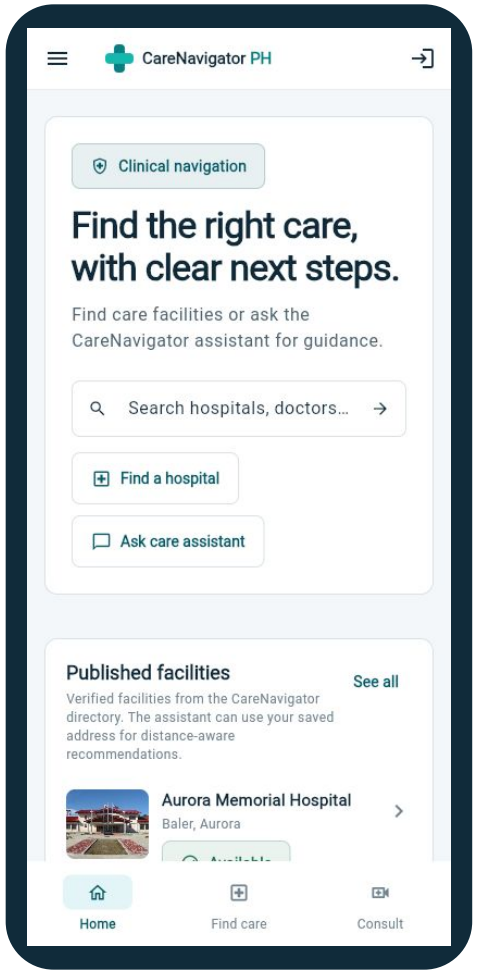
Source: SYSTEM_ARCHITECTURE.md section 6; README.md

SYSTEM DEMONSTRATION

The interface adapts from public discovery to role workspaces



Desktop public home - 1280 x 720

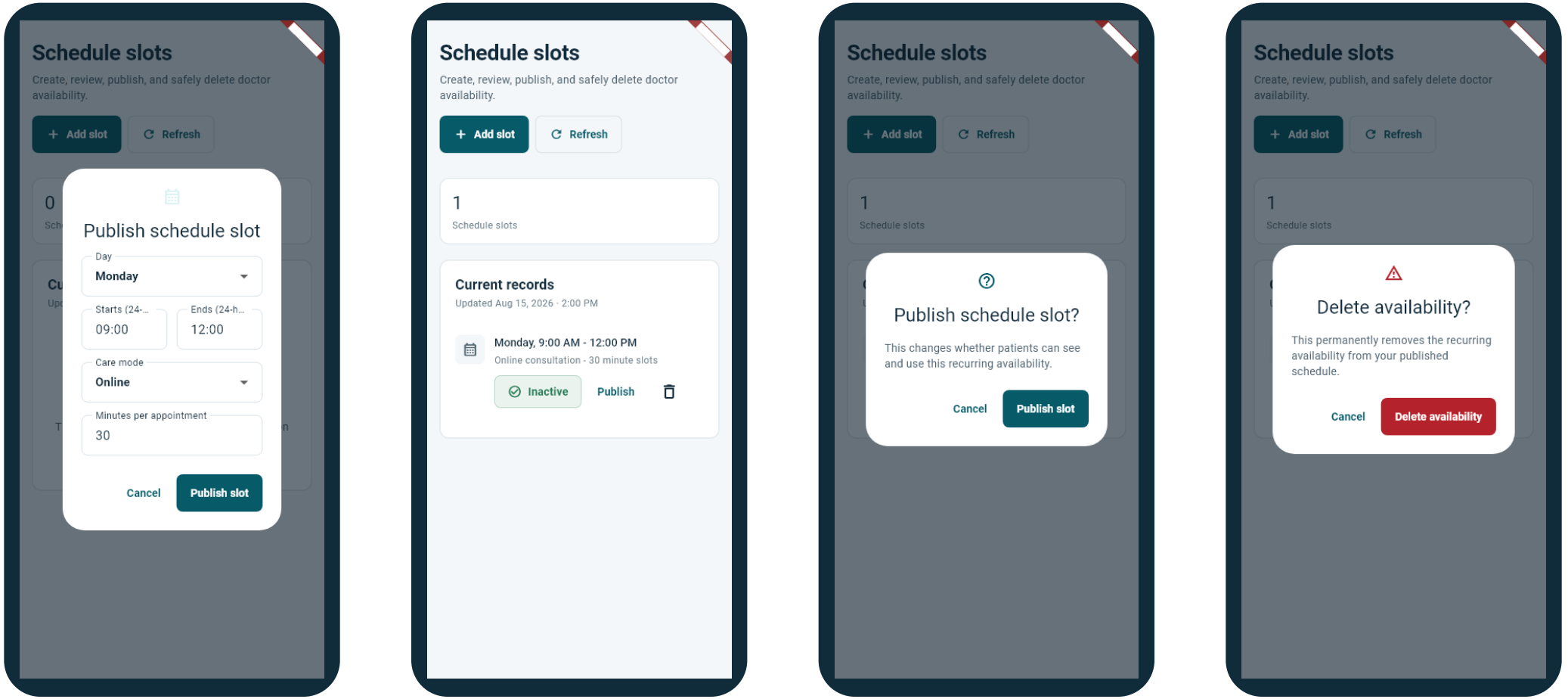


Mobile public home - 390 x 844

Source: submission_assets/live_app_home.png; live_app_home_mobile.png

RESPONSIVE IMPLEMENTATION

Mobile CRUD controls are visible at a real 390 x 844 viewport



Create | Read | Update | Delete

Schedule-slot workflow: publish, review, change visibility, and safely delete recurring availability.

Source: submission_assets/mobile CRUD screenshots - 390 x 844

TECHNICAL DESIGN

Layered architecture separates client, data, and privileged work

Flutter client — Adaptive mobile and web UI, GoRouter navigation, shared clinical design system

↓

State and access — Riverpod providers and typed Dart repositories as the only Supabase boundary

↓

Supabase platform — Auth, PostgreSQL, RLS, Realtime, private Storage, RPCs, triggers, Cron

↓

Server integrations — Edge Functions connect Groq, Gmail SMTP, OpenStreetMap, and OSRM

Source: SYSTEM_ARCHITECTURE.md sections 2-5

RISK CONTROLS

Clinical safety and authorization are enforced before convenience

Clinical safeguards

- Emergency warning signs trigger deterministic escalation.
- AI output is preliminary and requires licensed-doctor review.
- Medical-result processing tracks review and decision states.
- No unavailable or invented records are substituted in the UI.

Data safeguards

- RLS prevents cross-patient, cross-doctor, and cross-hospital access.
- Private files use short-lived signed URLs after authorization.
- Privileged workflows validate roles before server-side credentials are used.
- Sensitive administrative and workflow actions are audited.

Source: SYSTEM_ARCHITECTURE.md sections 7-8; VERIFICATION_REPORT.md

IMPLEMENTATION EVIDENCE

The backend contract is broad, live, and access-controlled

49

RLS-enabled public tables

19

Realtime-published tables

5

Active Edge Functions

Private clinical Storage, audited RPCs, scheduled notification dispatch, and server-side secrets complete the platform boundary.

Source: README.md - Supabase; PRE_BUILD_REPORT.md - Live Supabase contract inventory

VERIFICATION RESULTS

Core quality gates passed for code, tests, and builds

Flutter static analysis	PASSED
Automated Flutter tests	26 PASSED
Flutter web release build	PASSED
Android debug build	PASSED
Live migration verification	PASSED
Ordinary live-contract test	1 INTENTIONAL SKIP

Scope note: live seeded-account workflows require configured non-production accounts and relationship data.

Source: VERIFICATION_REPORT.md - Verification summary and runtime-tested behavior

LIMITATIONS AND NEXT STEPS

Remaining work centers on production validation

1. **Connected user testing**
   Exercise patient, doctor, hospital-admin, and super-admin sessions with dedicated test accounts.

2. **External providers**
   Validate SMTP delivery, Groq responses, online consultation channels, and scheduled dispatch.

3. **Device coverage**
   Complete physical Android testing and an iOS build and device pass on macOS.

4. **Security hardening**
   Enable leaked-password protection and complete an independent privacy and healthcare compliance review.

5. **Operations**
   Add centralized observability, backup and recovery procedures, key rotation, and load testing.

Source: VERIFICATION_REPORT.md - Remaining production verification; SYSTEM_ARCHITECTURE.md section 11

Defense conclusion

CareNavigator PH demonstrates that one responsive, role-aware system can connect care discovery, consultation workflows, protected clinical records, and hospital operations without weakening authorization boundaries.

What the defense establishes

- The system solves a coherent care-navigation problem.
- Its architecture separates UI, state, data access, and privileged services.
- Its safety, security, and verification evidence are documented and testable.

Ready for demonstration and questions

Source: README.md; SYSTEM_ARCHITECTURE.md; VERIFICATION_REPORT.md